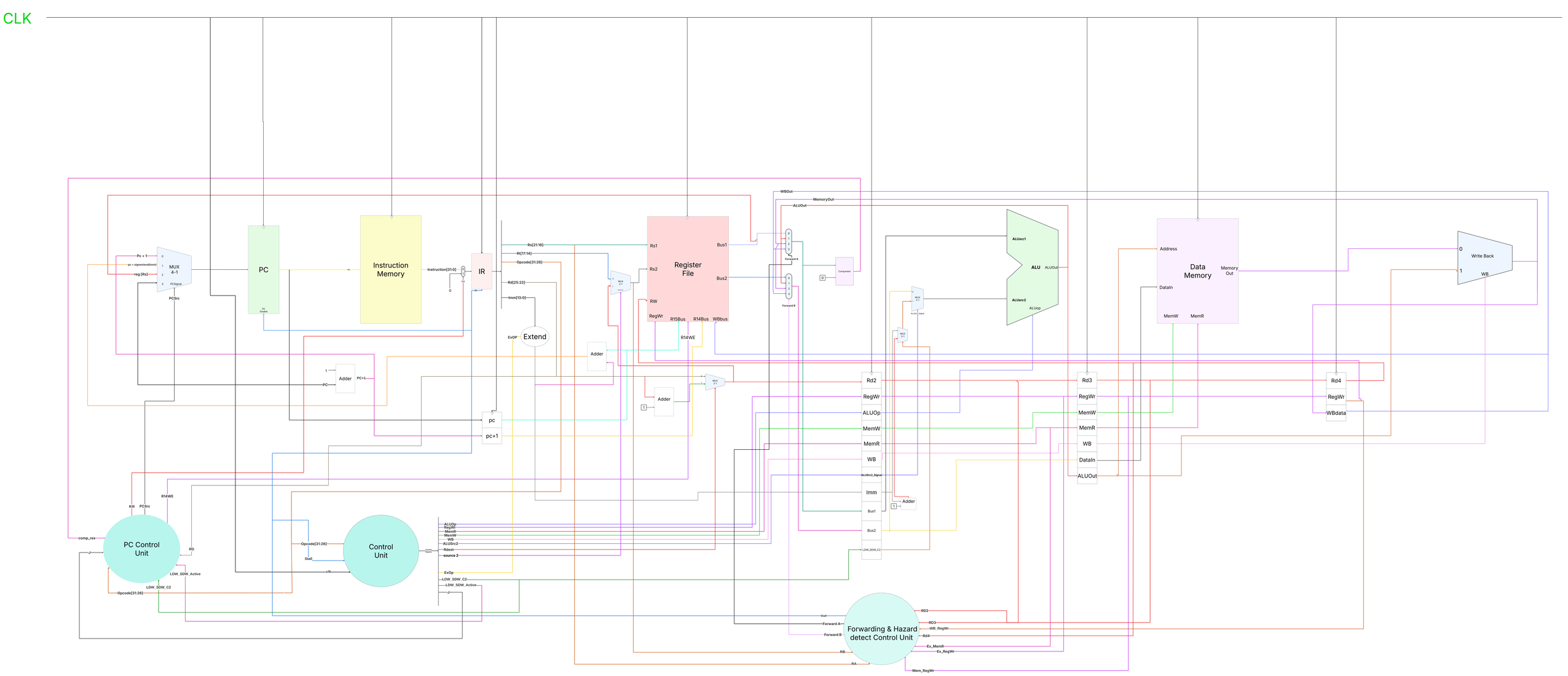
CLK
MUX 4-1
Pc + 1
pc + signextend(Imm)
reg [Rs]
PCSrc
PCSignal
PC
PC Disable
Instruction Memory
Instruction[31:0]
IR
Rs[21:18]
Rt[17:14]
Opcode[31:26]
Rd[25:22]
Imm[13:0]
ExOP
Extend
Register File
Rs1
Rs2
RW
RegWr
Bus1
Bus2
R15Bus
R14Bus
WBbus
R14WE
WBOut
MemoryOut
ALUOut
Forward A
Forward B
Comparator
ALUsrc1
ALU
ALUsrc2
ALUop
Address
Data Memory
DataIn
MemW
MemR
Memory Out
Write Back
WB
Adder
PC+1
pc
pc+1
Rd2
RegWr
ALUOp
MemW
MemR
WB
ALUSrc2_Signal
Imm
Bus1
Bus2
LDW_SDW_C2
Rd3
RegWr
MemW
MemR
WB
DataIn
ALUOut
Rd4
RegWr
WBdata
Kill
PCSrc
R14WE
comp_res
J
RD
LDW_SDW_Active
LDW_SDW_C2
Opcode[31:26]
PC Control Unit
Control Unit
Stall
clk
ALUOp
RegWr
MemW
WB
ALUSrc2
RdSrc
source 2
ExOp
LDW_SDW_C2
LDW_SDW_Active
J
Forwarding & Hazard detect Control Unit
Stall
Forward A
Forward B
RB
RA
RD2
RD3
WB_RegWr
Rd4
Ex_MemB
Ex_RegWr
Mem_RegWr